

Dr. SUMEGH S. THAREWAL (PhD, Post-doc, SMIEEE)

Associate Professor and Program Head of MCA & MTech

School of Advance Computing,

DBS Global University, Dehradun, Uttarakhand, India.

Adjunct Faculty, ICARUS AI, California, USA

Adjunct Faculty, Samarkand State University, Samarkand, Uzbekistan

Ex-Research Consultant, Market Insight Solutions, Pune

Former Assistant Professor, Symbiosis International University, Pune, India.

Former Program Head of M.Sc. Blockchain Technology, MIT WPU, Pune.

Former Fellow UGC SAP BSR, Government of India.

Former Fellow Avishkar, Government of Maharashtra.

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Permanent Address: Flat no R602, Gajendra Vihar,
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Uttarakhand, India

Personal Information

Date of birth: July 15, 1988
Phone: +91 -9503867301, +91-8669134570
Passport No: AI252405

Desirable Position: Professor Research/ Teaching/ any Leadership Role

Job Objective: To obtain a responsible position where my teaching and research experience will be helpful with an opportunity for continuous growth.

Highlights of Qualifications:

- ✚ Hands-on experience at the graduation and post-graduation level education facility.
- ✚ Familiarity working at a College and University department.
- ✚ Outstanding ability to handle syllabus at the undergraduate and graduate curriculum.
- ✚ Remarkable ability to teach, inspire and develop young people.
- ✚ Wide experience in providing assistance to Head Professor.
- ✚ Thorough knowledge of web-based instructional technologies
- ✚ Sound knowledge of teaching methodologies
- ✚ Familiarity with scripting and computer programming operations
- ✚ Good understanding of honors, advanced degrees, and awards in the subject field

- ✚ Ability to prepare and submit grant applications.
- ✚ Ability to maintain a good report with students and faculty staff.
- ✚ Ability to teach students with diverse backgrounds.

Education

2023- 2024	Postdoctoral Researcher , Free University of Bozen- Bolzano, Italy.
2014 -2020	Ph.D. in Computer Science and Technology Topic Entitled “ Multimodal Biometric Human Recognition System using 3D Face and 3D Ear ”, Department Computer Science and Information Technology, Dr. Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra, India.
2009-2011	M.Sc.in Computer Science , MGM’S Dr G Y Pathrikar College of Computer Science & Information Technology N-6 CIDCO, Aurangabad 431001, Maharashtra, India.
2005-2009	B.Sc.in Science , Yeshwantrao Chavan College, Sillod, Dist.: - Aurangabad, Maharashtra, India.
2003-2005	H.S.C , Yeshwantrao Chavan College Sillod, Dist.: - Aurangabad, Maharashtra, India.
2002-2003	S.S.C , New High School Bhokardan Dist.: - Jalna, Maharashtra, India.

Research Experience

1. Worked as Postdoctoral researcher on project entitled "**Robot-assisted assembly and disassembly to facilitate the remanufacturing and reuse of products (REMANUFACTURING) + Structured Representation Learning for Vision**", at **SMACT Lab NOI tech Park, Free University of Bozen-Bolzano, Italy.**

❖ Duties and Responsibilities:

- ✚ To develop advance perception enabling human-robot collaboration.
- ✚ Design and develop the Algorithm for human tracking and action recognition.
- ✚ To capture, record the video from fix cameras with video and sensor streams from VR glasses worn by human.
- ✚ Develop methods as perception components in human-robot collaboration demonstrator.
- ✚ Speedup the human-robot collaboration process, reduce the labor cost.

2. Worked as a **Senior Research Fellow (From 8 August 2013 – 13 October 2020)** in A Major Research Project UGC SAP- II, DRS Phase- II, and the Government of India sponsored project "**Multimodal Biometric System Development**".

Project Principal Investigator: Professor K V Kale, Department of Computer Science and Information Technology, Dr. Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra, India

❖ **Duties and Responsibilities:**

- ✚ To independently undertake advanced research in the relevant discipline area using, or developing, the appropriate methods, and techniques, managing associated staff, and contributing to the project.
- ✚ To Design and Develop a Database of Multimodal Biometric System using fingerprint, face, iris, signature, voice, ear, palm print, dorsal hand, nail, knuckle, 3D face, 3D ear, Video, EEG, HRV, ECG, Hyperspectral face biometric modalities. Study the Standards available Database like NIST, IEEE.
- ✚ Design and develop our own Standards of Database.
- ✚ Design and develop the new algorithm for multimodal biometric recognition.
- ✚ Design, develop and update the Lab Presentation.
- ✚ Give the detailed Presentation, sensor information, and practical demonstration to the lab visitors.
- ✚ To lead the production of high-quality research reports, and publications as required by the funding body or for dissemination to the wider academic community.
- ✚ To take lead responsibility for resolving problems affecting the delivery of research projects and for planning and implementing future research projects that emerge from the current project.
- ✚ To lead and promote activities designed to develop collaborative research with research colleagues and support staff internally and to actively develop appropriate external contacts and networks relevant future collaboration and funding opportunities.
- ✚ Write and publish quality research paper in peer reviewed journals.
- ✚ Report the work progress of project to the principal investigator of project.
- ✚ To maintain and update the Software, Contingency record, Lab report and Lab register.
- ✚ To lead relevant meetings associated with the research project or related activities.
- ✚ To contribute to the teaching profile of the department.

❖ Sensors and scanner handling during the Project

- ✚ NITGEN Finger key Hamster (Fingerprint scanner)
- ✚ FutronicFS88(fingerprint scanner)
- ✚ CrossMatchLScan500P (Palm print scanner, multi finger, rolled finger scanner)
- ✚ Cross match I Scan 2 (Iris Scanner)
- ✚ Iball signature Pad (Signature acquisition)
- ✚ 3D Sense scanner (3D face and 3D ear scanner)
- ✚ Microsoft Kinect 360(3D face and 3D ear scanner)
- ✚ Heart Rate variability device (HRV data acquisition)
- ✚ Mind wave Mobile (EEG data acquisition)
- ✚ ECG Machine (ECG data acquisition)
- ✚ D-EYE Portable Retinal Imaging Scanner (Retina data acquisition)
- ✚ Canon EOS 700 DSLR, Intex-IT305WC, iBallUSB2.05g lens (2D Face and 2D Ear ,3D Face and Ear data acquisition)

Research Area of Interest

- ✚ Biometrics, Blockchain Technology, Machine Learning, Artificial Intelligence, Pattern Recognition, Image Processing, Computer Vision, Deep Learning, Internet of Things (IoT)

Patents- (Total Number: - 15)

- 1) **Patent Authority: - Government of India**
Patent Application Number: - 202131002165
Title of Invention: - MACHINE-LEARNING BASED METHOD FOR NETWORK SECURITY FOR VIRTUAL REALITY COMMUNICATION NETWORK
Current Status: - Published
- 2) **Patent Authority: - Australian Government**
Patent Application Number: - 2021101577
Title of Invention: - AN IOT-BASED SYSTEM FOR EMERGENCY HEALTHCARE
Current Status: - Granted
Date of File: - 27 March 2021
Date Grant: - 05 May 2021
- 3) **Patent Authority: - Australian Government**
Patent Application Number: - 2021102972
Title of Invention: AN IMAGE PROCESSING AND DEEP LEARNING-BASED SYSTEM FOR ADVERTISEMENT
Current Status: Granted

Date of File: - 30 May 2021
Date Grant: - 09 March 2022

4) Patent Authority: - Australian Government

Patent Application Number: 2021103135

Title of Invention: AN ARTIFICIAL INTELLIGENCE, IOT AND DEEP LEARNING-BASED SYSTEM FOR DETECTION OF COVID-19, BLACK FUNGUS, AND WHITE FUNGUS DISEASE

Current Status: Granted

Date of File: - 05 June 2021

Date Grant: - 16 March 2022

5) Patent Authority: - Australian Government

Patent Application Number: 2021103317

Title of Invention: A SYSTEM FOR BLOCKCHAIN LINKED INTERNET OF THINGS-BASED RAILWAY DIGITAL DISPLAY READER

Current Status: Granted

Date of File: - 12 June 2021

Date Grant: - 9 March 2022

6) Patent Authority: - Australian Government

Patent Application Number: 2021103730

Title of Invention: AN IOT AND CLOUD COMPUTING ENABLED 5G SENSOR NETWORK FOR SMART CITY IMPLEMENTS

Current Status: Granted.

Date of File: - 29 June 2021

Date Grant: - 04 August 2021

7) Patent Authority: - Government of India

Patent Application Number: 202111043586

Title of Invention: IOT ENABLED METHOD FOR DETECTING IMPRESSIONS BASED ON ARTIFICIAL INTELLIGENCE AND DEEP LEARNING

Current Status: Published

8) Patent Authority: - Government of India

Patent Application Number: 202141050781

Title of Invention: A BLOCK CHAIN-BASED SYSTEM AND METHOD FOR PERSONAL IDENTITY VERIFICATION

Current Status: Published

9) Patent Authority: - Government of South Africa

Patent Application Number: 2021/10723

Title of Invention: AN ARTIFICIAL INTELLIGENCE BASED DEVICE TO CONTROL A SWITCHGEAR USING WIRELESS INTERFACE CIRCUIT

Current Status: Granted
Date of File: - 23 October 2021
Date Grant: - 29 June 2022

- 10) **Patent Authority: - Government of Germany**
Patent Application Number: G11450DE
Title of Invention: AN IOT AND MACHINE LEARNING-BASED PREDICTION AND EARLY WARNING DEVICE FOR PATIENTS WITH MENTAL DISORDERS
Current Status: Granted
Date of File: - 1 April 2022
Date Grant: - 27 May 2022
- 11) **Patent Authority: - Government of Germany**
Patent Application Number: G11451DE
Title of Invention: IOT-BASED COVID-19 EARLY WARNING SYSTEM FOR SENIOR CITIZENS
Current Status: Granted
Date of File: - 2 April 2022
Date Grant: - 2 June 2022
- 12) **Patent Authority: - Government of India**
Patent Application Number: 368826-001
Title of Invention: ARTIFICIAL INTELLIGENCE POWERED ROBOT FOR HOME ASSISTANCE
Current Status: Granted
- 13) **Patent Authority: - Government of India**
Patent Application Number: 202221071202
Title of Invention: METHOD OF OBJECT DETECTION AND TRAFFIC CONTROL SYSTEM BASED ON ARTIFICIAL INTELLIGENCE
Current Status: Granted
- 14) **Patent Authority: - Government of UK**
Patent Application Number: 6276267
Title of Invention: BIOMETRIC ATTENDANCE DEVICE WITH TOUCH FREE BODY TEMPERATURE SCREENING
Current Status: Published
- 15) **Patent Authority: - Government of India**
Patent Application Number: 202311031058
Title of Invention: AGRICULTURAL ROVER SYSTEM FOR AUTOMATIC PEST CONTROL
Current Status: Published

Research Profile

- ✚ **Patents: - 15** (5 Australian, 2 German, 1 UK, 1 South African, 6 Indian)
- ✚ **Scopus Index published research papers: - 41**
(6 research papers are in pipeline and publish very soon)
Scopus profile link: - <https://www.scopus.com/authid/detail.uri?authorId=56943472800>)
- ✚ **SCI published research Papers: - 10** (2 research papers are in pipeline and publish very soon)
- ✚ **Books: - 7**
- ✚ **Book Chapters: -6**
- ✚ **Fellowship Received: - 4** (2 International, 1 National, 1 State level)
- ✚ **Google Scholar**
Cited by 692
h-index 13
i10-index 15
URL: <https://scholar.google.co.in/citations?user=1dfdWf8AAAAJ&hl=en>
- ✚ **Research Gate**
Research Interest Score 503.2
Citation 603
h-index 12
Reads 53,320
URL: https://www.researchgate.net/profile/Sumegh_Tharewal3
- ✚ **Scopus Author ID: 56943472800**
- ✚ **ORCiD ID: - 0000-0002-5130-7358**
- ✚ **Web of Science Researcher (Publons) ID: - AAD-7897-2020**

Projects on Going

- ✚ Design and Development of Blockchain-Based Robust Multimodal Biometric System Using 3D Face and 3D Ear.
- ✚ Exploring the Use of AI Chabot for Self-Help Depression Interventions among School Students
- ✚ An efficient alzheimer's disease prediction system with phenotype generation and six-class classification
- ✚ Design and Development of AI Based Real Time Fingerprint Authentication System

Projects Successfully Completed

1. Robot-assisted assembly and disassembly to facilitate the remanufacturing and reuse of products (REMANUFACTURING) + Structured Representation Learning for Vision
2. Multimodal Biometric Human Recognition system using 3D Face and 3D Ear
3. Design and Development of Deep Learning Architecture for Classification, Recognition, and object Detection from Multimodal Biometric Database.
4. Face, Fingerprint, signature, and iris-based multimodal biometric system for person Identification.
5. Face, signature, and Ear based Multimodal biometric system for human authentication.
6. Heart Rate Variability-based Multimodal Biometric System for Human Authentication and Identification.
7. Design and Development of a Low-cost 3D Scanner
8. Human heart Disease Prediction and classification using Machine Learning Approach.

Ph.D. Student Research Project

- ✚ Design and Development of Blockchain-Based Robust Multimodal Biometric System Using 3D Face and 3D Ear.
- ✚ An efficient alzheimer's disease prediction system using era-dpinn with phenotype generation and six-class classification
- ✚ Exploring the Use of AI Chabot for Self-Help Depression Interventions among School Students
- ✚ Early Detection of Kidney Disease Using Machine Learning Techniques
- ✚ Design and Development of AI Based Real Time Fingerprint Authentication System

Postdoctoral Student Research Project (Singapore Institute of Technology)

- ✚ A Study on Identifying and Controlling Digital Biases in Trading and Investments with Machine Learning Techniques
- ✚ Machine Learning Approaches to Battery Performance Prediction and Management in Rugged Tropics
- ✚ Predictive Battery Maintenance for Rugged Systems in Tropical Regions with Machine Learning

PG Student Research Projects

- ✚ Blockchain and Smart Contracts based system for Infrastructure.
- ✚ Role of Blockchain in Smart City
- ✚ Blockchain Technology for Beverage Industry
- ✚ NFT for Product Verification System
- ✚ Online Art Gallery using Blockchain Technology
- ✚ Smart Accident Prevention System using Artificial Intelligence
- ✚ Fake News Detector.
- ✚ Emotion Detection from Facial Expressions
- ✚ Forecasting Future Sales for a Retail Store Chain.
- ✚ MM Detection – Helmet Detection.
- ✚ Credit card fraud detection
- ✚ Supply Chain Management System for Apparel Industry
- ✚ Loan Eligibility Prediction System
- ✚ Iris Flower Classification
- ✚ Student Performance Prediction
- ✚ Mental health detection system
- ✚ House price prediction system
- ✚ Electricity Bill Reduction system
- ✚ Mobile Applications Security Improvements System
- ✚ A decentralized document sharing system.
- ✚ Gym Management System
- ✚ Electronic Health Record Management System using Blockchain Technology
- ✚ Fake Product Identification

- ✚ Secure Voting System
- ✚ Crowdfunding System
- ✚ Gold Mining Management System
- ✚ Blockchain In Real estate
- ✚ Decentralized NGO
- ✚ Decentralized KYC System
- ✚ Patient Data Management System
- ✚ Blockchain In Supply Chain
- ✚ Skill Verification System
- ✚ Vehicle rental system
- ✚ PageRank Algorithm
- ✚ Decentralized Lottery Platform
- ✚ IOT-Based Air Pollution Monitoring & Controlling System
- ✚ Blockchain-based Solution for Agri. & Food Supply Chain

Countries Visited During the Research Work

- ✚ Italy, France, Germany, Switzerland, Spain, Austria, Turkey, UAE, Uzbekistan

Teaching Experience: 14 Years and 6 Months

1. Designation: - Associate Professor and Program Head of MCA & MTech

Duration: - 8, August 2024 to till date

Organization: - DBS Global University, Dehradun, Uttarakhand, India.

❖ Duties and Responsibilities:

- ✚ International Program Coordinator
- ✚ Teaching and Research
- ✚ Member of Academic Council
- ✚ Member of Anti ragging Squad

- ✚ PhD Research Supervisor
- ✚ Member secretary of DRC (Departmental Research Committee)
- ✚ Member secretary of SRC (School Research Committee)
- ✚ Member of startup and Incubation
- ✚ Member of Discipline committee
- ✚ Organize the International Conference
- ✚ Publish Research papers in peer-reviewed journals.
- ✚ Apply for Research Projects.
- ✚ File Patents and copyrights.

2. Designation: - Postdoctoral Researcher

Duration: - 1 October 2023 to 26/7/2024

Organization: - Free University of Bozen- Bolzano, Italy.

❖ Duties and Responsibilities:

- ✚ Conduct independent research and contribute to the advancement of knowledge in the field of research project.
- ✚ work on a specific research project under the guidance of a principal investigator (PI).
- ✚ Design and conduct experiments, analyze data, write publications, and present their findings at conferences and journals.
- ✚ Discussions and collaborations with other researchers in same field.
- ✚ Gaining new research skills, contributing to the effort of the lab
- ✚ Develop the research skills, learn about the broader scientific community.

3. Designation: - Assistant Professor and Program Head of M.Sc. CA

Duration: - From 20 April 2023 to 25 September 2023

Organization: - Symbiosis Institute of Computer Studies and Research,
Symbiosis International University, Pune, India.

❖ Duties and Responsibilities:

- ✚ Teaching and Research
- ✚ Publish Research papers in peer-reviewed journals.
- ✚ Apply for Research Projects.

- ✚ File Patents and copyrights.

4. Designation: - Adjunct Faculty

Duration: - From 1, January 2023 to till date

Organization: - ICARUS AI E-Learning, Santa Monica, CA, USA

❖ Duties and Responsibilities:

- ✚ Developing, delivering, and reviewing course material and syllabi
- ✚ Creating a classroom environment where students feel comfortable participating and learning.
- ✚ Setting up assessments based on departmental regulations and requirements.

5. Designation: - Assistant Professor and Program Head of M.Sc. Blockchain Technology.

Duration: - From 13 January 2022 to 15 April 2023

Organization: Dr. Vishwanath Karad MIT World Peace University, Pune, India.

❖ Duties and Responsibilities:

- ✚ Plan, schedule, organize, coordinate, and monitor Lectures and Practical for the students.
- ✚ Complete the syllabus for the subjects assigned.
- ✚ Prepare Lecture Plan/ Lecture Materials/ Course Material
- ✚ Conduct at least three internal Tests during the year in the subjects assigned.
- ✚ Encourage students to participate in co-curricular and extra-curricular activities.
- ✚ Plan, deliver, and evaluate theoretical/practical instructions.
- ✚ Guide the students in the performance of practical tasks and skill exercises and evaluate their performance, Advise, and assist the students in their project works.
- ✚ Participate in professional development activities.
- ✚ Publish Research papers in peer-reviewed journals.
- ✚ Apply for Research Projects.
- ✚ File Patents and copyrights.
- ✚ Collaborations with reputed Universities and Sign the MoU with reputed Institutions.

6. Designation: - Assistant Professor and Program Head of MCA

Duration: - From 8 December 2020 to 7 January 2022.

Organization: Manipal University Jaipur, Rajasthan, India

❖ Teaching Duties and Responsibilities:

-  Teaching both core and elective courses in the field of specialization as allocated by the Head of the Department for various programs offered by the University.
-  Providing proper guidance and supervision of the project work undertaken by students and developing proper rapport with the industry/organization if the project is industry related.
-  Making the teaching more effective and interesting to the students using multimedia teaching aids.
-  Contributing to curriculum, resource, program and subject design, development, management, and review as required to ensure that learning and teaching in the faculty and across disciplines reflects best practice and a command of the field.
-  Continuously improving own teaching and learning, including developing a comprehensive teaching portfolio, and contributing to the continuous development and improvement of learning and teaching across the university.
-  Participating in scholarly activities that influence and enhance learning and teaching in the University.
-  Preparing high quality subject delivery and learning support materials using any web-based platforms, electronic library information systems and other teaching and learning systems developed for use in the University.
-  Acting as a subject coordinator and/or area or discipline coordinator including managing the area/discipline including providing feedback and contributing to performance management; supervising tutors; carrying out related planning and coordination responsibilities; organizing the preparation and marking of assignments and all examinations; invigilating examinations; and submitting grades.
-  Conducting lectures, tutorials, workshops, practical classes, demonstrations, field excursions, clinical sessions and other appropriate learning activities as required.

- ✚ Obtaining feedback from students, peers and employers on individual teaching, the subjects, and programs for which they are responsible, and the implementation and pedagogy.
- ✚ Contributing to internal and external reviews and accreditation of subjects and programs for which they are responsible.
- ✚ Providing respect and support for the development of students as individuals and thereby contributing to creating the unique Bond experience for students in their learning. Providing approaches to learning that influence, motivate, inspire students to learn.

❖ **Research Duties and Responsibilities**

- ✚ Providing a continuing high level of personal commitment to, and achievement in, a particular scholarly area and as a result, generating high-level research outcomes
- ✚ Developing a record of publication in top-ranked and other research and scholarly journals and other forms of publication and recognition of that record within the research community.
- ✚ Participating in research projects and research teams (where appropriate)
- ✚ Succeeding in applications for research funding, whether individually or as part of a team
- ✚ Supervising the program of study for honors students and postgraduate students undertaking research projects
- ✚ Participating in development opportunities to improve research and supervision skills.
- ✚ Contributing to developing, mentoring, and providing feedback to junior researchers
- ✚ Contributing to the Faculty and University in the policy development, management, and review of research

Departmental Academic Roles and Responsibilities

- ✚ Exam Co-Coordinator
- ✚ Startup and Incubation Coordinator
- ✚ Internationalization
- ✚ Research Guide
- ✚ Monthly Review Meeting Presentation

7. Designation: - Teaching Assistant

Duration: - From 12 August 2013 – 12 July 2018.

Organization: - Department of Computer Science and Information Technology,
Dr. Babasaheb Ambedkar, Marathwada University, Aurangabad, Maharashtra, India.

❖ Duties and Responsibilities:

- ✚ Assist teachers with lesson preparation by getting materials ready and setting up equipment.
- ✚ Work with the lead teacher to monitor the class schedule.
- ✚ Revise lesson material with students individually or in small groups.
- ✚ Collaborate with lead teachers to recognize issues students are facing and recommend solutions.
- ✚ Help lead teachers to create lesson plans.
- ✚ Attend all training classes, parent conferences, and faculty meetings.
- ✚ Help teachers with recordkeeping, such as taking attendance and calculating grades.

8. Designation: - Visiting Assistant Professor

Duration: - From 1 August 2018 to 30 April 2019

Organization: -Government Institute of Forensic Science Aurangabad.

❖ Duties and Responsibilities:

- ✚ Instructing on established courses.
- ✚ Conceiving, refining, and teaching novel lecture series.
- ✚ Conducting laboratory courses, tutorials and seminars of the programs assigned to in an effective manner, to improve the practical knowledge of the student.
- ✚ Providing proper guidance and supervision of the project work undertaken by students and development of proper rapport with the industry/organization if the project is industry related one.
- ✚ Making the teaching more effective and interesting to the students using multi- media teaching aids.
- ✚ Setting and examining coursework and summative assessments.
- ✚ Attending faculty-organized training and networking meetups.

9. **Designation: - Senior Software Developer**

Duration: - From 20 September 2018 to 22 October 2020.

Organization: - Inficom Solutions Pvt. Ltd, Aurangabad.

❖ **Duties and Responsibilities:**

- ✚ Researching, designing, implementing, and managing software programs.
- ✚ Testing and evaluating new programs.
- ✚ Identifying areas for modification in existing programs and subsequently developing these modifications.
- ✚ Writing and implementing efficient code.
- ✚ Determining operational practicality.
- ✚ Developing quality assurance procedures.
- ✚ Deploy software tools, processes, and metrics.
- ✚ Maintaining and upgrading existing systems.
- ✚ Training users.
- ✚ Working closely with other developers.

10. **Designation: - Assistant Professor**

Duration: - From 26 July 2011 to 31 July 2013

Organization: - Vivekanand College Aurangabad.

❖ **Duties and Responsibilities:**

- ✚ Assisting with various departmental duties and providing academic support to Professors and other staff.
- ✚ Teaching and supervising undergraduate and graduate students.
- ✚ Providing demonstrations and supervising experiments and investigations.
- ✚ Providing Professors and Department Heads with feedback on student progress.
- ✚ Attending faculty and departmental meetings and voicing concerns or providing suggestions for improvement.
- ✚ Conducting research and publishing papers in academic journals.
- ✚ Answering questions in class or via email or telephone.

Subjects Taught

- ✚ M.Sc. Blockchain Technology (Blockchain Ecosystem, Advance Data Structure, Blockchain Architecture, Network Security, and API Interfacing)
- ✚ M.sc Digital Cyber Forensic (Python, MATLAB, Research Methodology, Biometrics)
- ✚ Post-Graduation Diploma in Digital Cyber Forensic
- ✚ M.C.A (Machine Learning, Artificial Intelligence, Digital Image Processing, Operating System, Computer Networks, Cryptography and Network Security)
- ✚ MTech (Research Methodology)
- ✚ MBA (Information Security)
- ✚ M.Sc. CA (Fundamentals of Computer Networking)
- ✚ B.sc Computer Science (Operating System, C, C++, Computer Network)
- ✚ B.sc Digital Cyber Forensic (C, C++, Python)
- ✚ BCA (Fog Computing, Artificial Intelligence, Software Engineering, Computational Thinking and UML, Python)
- ✚ B.com (Computer Application)
- ✚ E-commerce, ERP, and E-banking

Awards and Achievements

1. Got **Research Excellence Award 2019** by Institute of Scholars Bangalore India.
2. Got Indo Asian Young Scientist Award 2021 in Biometrics by International Multidisciplinary Research Foundation (IMRF) Registered with Government of India NITI Aayog NGO Darpan.
3. Got Selected for **UGC BSR Fellowship** Government of India New Delhi from 2013 2018 (2 Year as Junior Research Fellow and 3 years as Senior Research Fellow).
4. Got **first prize** in university level Avishkar 2014 under Engineering and technology category organized by Dr. BAMU Aurangabad on November 24-25, 2014.
5. Got **second prize** in university level Avishkar 2017 under Engineering and technology category organized by Dr. BAMU Aurangabad on December 23-25, 2017.
6. Awarded **second prize** at Post PG level under the discipline Engineering and technology in the 12 Maharashtra state inter-university research convention Avishkar 2017 held at

- Mahatma Phule Krishi Vidyapeeth, Rahuri during January 15-17, 2018.
7. Presented One research Paper and one Poster in 15 International Summer School for Advanced Studies on “**Assuring Trustworthiness of Biometrics**” held at hotel EL Faro in Alghero (ITALY) during 11-15 June 2018 organized by University of Sassari in co-operation with the Michigan State University and The University of Halmstad and Technically Co-sponsored by the International Association for Pattern Recognition(IAPR),THE European Association for Signal Processing(EURASIP) and by the Institute of Electrical and Electronics Engineers(IEEE).
 8. Got selected for **Avishkar Research Fellowship by Government of Maharashtra** (July 2018-June 2020).
 9. Elected **Sectional committee member** of ICT section of Indian Science Congress Association (2019-2020)
 10. Certificate of Appreciation for **Excellent Contributions in the Academics, Research & Development Activities** at the School of Basic Sciences, Faculty of Science Manipal University Jaipur Awarded this day, 26th of March 2021.
 11. Received **Senior member IEEE** Recognition in August 2023.

National Conference Publication

1. Krishna Shinde, **Sumegh Tharewal**, Charansing Kayte, “Person Identification based on Offline Signature using Deep Learning”, 3rd National Students' Research Conference, 28 Feb 2022, MIT World Peace University, Pune.
2. **Sumegh Tharewal**, Hanumant Gite, Siddharth Dabhade, K. V. Kale, “Design and Development of Standard 3D Face and 3D Ear Database: A Multisensory Approach”, National Conference on Convergence of Technology, 23–24 Feb 2018, Ahmednagar, MH.
3. Krishna Shinde, **Sumegh Tharewal**, “A Study of Multimodal Biometric Person Identification System using Face, Fingerprint and Iris”, National Conference on Convergence of Technology, 23–24 Feb 2018, Ahmednagar, MH
4. Bharat Naiknaware, **Sumegh Tharewal**, S. N. Deshmukh, “Tracking the Website Real-Time Activity using Google Analytics”, NCEITCS-2014, April 1–2, 2014, Hyderabad.
5. **Sumegh Tharewal**, Surekha Kalyankar, Ratnamala Selukar, Rita Patil, “Study of Embedded Systems”, Emerging Trends in Information Technology, 12 Jan 2011, Pune.

6. **Sumegh Tharewal**, Surekha Kalyankar, "Cloud Computing with Azure Platform", 7th National Conference on Advances in IT & Management, 13 Feb 2011, Chalisgaon, MH.

International Conference Publication

1. V. Musale, P. Bhosale, **S. Tharewal**, M. Rao, O. Ninawe and A. Amune, "Leveraging Blockchain for Secure Mobile Payment Systems: A Privacy Centric Approach," 2025 International Conference on Intelligent and Cloud Computing (ICoICC), Bhubaneswar, India, 2025, pp. 1-6, doi: 10.1109/ICoICC64033.2025.11052073. **(Scopus)**
2. P. Daga, S. Dumre, M. Rao, K. V. Kumar, **S. Tharewal** and D. Bhagat, "Sentiment Analysis of Hazardous Events and Disasters," 2025 International Conference on Intelligent and Cloud Computing (ICoICC), Bhubaneswar, India, 2025, pp. 1-6, doi: 10.1109/ICoICC64033.2025.11052003. **(Scopus)**
3. Qureshi, I., Kale, V., & **Tharewal, S.** (2025, June). Enhancing Multifactor Authentication with Machine Learning: A Comprehensive Framework for Robust User Verification. In International Conference on Advances and Applications in Artificial Intelligence (ICAAAI 2025) (pp. 646-653). Atlantis Press. **(Scopus)**
4. Goje, S. P., & **Tharewal, S.** (2024, February). Phishing in the Inbox: A Systematic Examination of Email-Based Cyber Threats. In NIELIT's International Conference on Communication, Electronics and Digital Technologies (pp. 355-364). Springer Nature Singapore. **(Scopus)**
5. V. Kaur, D. P. Bhatt, **S. Tharewal** and P. K. Tiwari, "A Decentralized Blockchain-Enabled Immutability Mechanism Integrated with Multimodal Biometric," UPCON 2024, IEEE, Lucknow, India, 2024, pp. 1-8. **(Scopus)**
6. Lahase, A., Bhable, **S., Tharewal, S.**, Chitte, S., & Nimbhore, S. (2024, September). Enhancing Automatic Speaker Diarization for Marathi and Hindi Languages. In International Conference on Smart Cyber Physical Systems (pp. 383-394). Springer Nature Singapore. **(Scopus)**
7. Misal, N., **Tharewal, S.**, Kanade, S. P., Lahase, A., Bhable, S. G., Dhepe, A., & Jain, P. (2024, September). Advancing Indian Sign Language Recognition (ISLR). In International Conference on Smart Cyber Physical Systems (pp. 695-708). Springer Nature Singapore. **(Scopus)**

8. Pande, V. K., Kale, V. K., & **Tharewal, S.** (2024, February). Audio data feature extraction for speaker diarization. In NIELIT's International Conference (pp. 243–255). Springer Nature Singapore. **(Scopus)**
9. Malche, T.; **Tharewal, S.**; Dhanaraj, R.K., “Automated Damage Detection on Concrete Structures Using Computer Vision and Drone Imagery”, Eng. Proc. 2023, 56, x. **(Scopus)**
10. V. Kaur, D. P. Bhatt, **S. Tharewal** and P. K. Tiwari, "Blockchain-based Secure Storage Model for Multimodal Biometrics Using 3D Face and Ear," InCACCT 2023, pp. 860–865. **(Scopus)**
11. Goje, S.P., Ansari, G.A., Ansari, M.D., **Tharewal, S.** (2023). Phishing Email Mitigation Using Back-Propagation Neural Network. In ICDSMLA 2022. Springer. **(Scopus)**
12. Radhika Thakkar, Bhumika Ostwal, Suraj Gandhi, Dhairya Shah, **Sumegh Tharewal**, "Diabetes Classification and Diet Recommendation", 4th National Student Research Conference, May 2023.
13. Timothy Malche, Pradeep Kumar Tiwari, **Sumegh Tharewal** and Rajesh Tiwari, “Smart Waste Management System for Smart City based on IoT”, ICICE-2020, Feb 11–13, 2021. **(Scopus)**
14. Kapila Pareek and **Sumegh Tharewal**, “A brief review on parameters of Link stability in MANETs”, ICCIPS, March 9–10, 2021. **(Scopus)**
15. Devershi Pallavi Bhatt, **Sumegh Tharewal**, Shilpa Sharma, Devanshi Ledwani, “A review on RFID Based Children Safety Management system”, RAMSACT, November 24–26, 2021. **(Scopus)**
16. Timothy Malche, Devershi Pallavi Bhatt, **Sumegh Tharewal**, “A Portable Water Monitoring Device for Smart City Based on IoT”, SUSCOM 2021, March 19–20, 2021. **(Scopus)**
17. Krishna Shinde, **Sumegh Tharewal**, Kiran S. Suryawanshi, Dr. Charansing N. Kayte, “Python Based Face Recognition...”, Virtual Conference, JNEC, Aurangabad, 30–31 Oct 2020. **(Scopus)**
18. Mohammed Waseem Ashfaque, **Sumegh Tharewal**, Dr. S Iqbal, Dr. C N Kayte, “A review on Intelligent Tutoring System (Chatbot)”, Virtual Conference, JNEC, Aurangabad, 30–31 Oct 2020. **(Scopus)**

19. Siddharth B. Dabhade, Nagsen Bansod, Yogesh S Rode, M M Kazi, **Sumegh Tharewal**, K V Kale, “Hyperspectral Image Analysis for Human Authentication”, ICIIP 2017, DOI: 10.1109/ICIIP.2017.8313729. **(Scopus)**
20. **Sumegh Tharewal**, Hanumant Gite, K V Kale, “3D Face and 3D Ear Recognition: Process and Techniques”, ICCTCEEC-2017, Mysuru, 8–9 Sept 2017. **(Scopus)**
21. **Sumegh Tharewal**, K V Kale, “A Review on Multimodal Biometric Human Recognition Using 3D Face and 3D Ear”, ICKE-2016, Aurangabad, 21–23 Dec 2016.
22. N. Akhter, **S. Tharewal**, V. Kale, A. Bhalerao, K. V Kale, “Heart-Based Biometrics and HRV”, International Doctoral Symposium, University of Calcutta, May 23–25, 2015. **(Scopus)**
23. Nazneen Akhter, **Sumegh Tharewal**, Hanumant Gite, KV Kale, “Microcontroller Based R-R Interval Measurement using PPG Signals”, ICACCI’15, Kochi, 10–13 Aug 2015. **(Scopus)**
24. Nazneen Akhter, Hanumant Gite, **Sumegh Tharewal**, KV Kale, “Computer Based RR-Interval Detection System...”, ICACCI’15, Kochi, 10–13 Aug 2015. **(Scopus)**
25. **Sumegh Tharewal**, Hanumant Gite, Suvernsing Bhable, Santosh Maher, K V Kale, “Development of Multimodal Biometric Human Recognition System using 3D Face and 3D Ear”, EECCMC, 28–29 Jan 2018.
26. Vijay Kale, **Sumegh Tharewal**, Prapti Deshmukh, “Development of Voice Biometric System for Person Identification”, EECCMC, 28–29 Jan 2018. **(Scopus)**
27. Ganesh Ragade, **Sumegh Tharewal**, Bharti Gawali, “Development of Leakage Detection System in Water Distribution System: A review”, EECCMC, 28–29 Jan 2018. **(Scopus)**

International Journal Publication

1. Kaur, V., Bhatt, D.P., Tharewal, S. et al. Intelligent multimodal 3D biometric recognition using PointNet++ for robust face-ear authentication. Sci Rep 15, 43793 (2025). <https://doi.org/10.1038/s41598-025-27757-5> , Scientific Reports by Nature **(SCI Q1)**
2. Dutta, M., Gupta, D., **Tharewal, S.**, Goyal, D., Sandhu, J. K., Kaur, M., ... & Alanazi, J. M. (2025). Internet of Things-based smart precision farming in soilless agriculture: Opportunities and challenges for global food security. IEEE Access. **(SCI Q1)**

3. Banerjee, D., Krishnan, S. B., Upreti, K., **Tharewal, S. S.**, Shankar, U., Kshirsagar, P., Kumar, M. (2025). Data-Driven Drug Discovery Optimization for Breast Cancer Using Interpretable Machine Learning Models. *Journal of Visualized Experiments (JoVE)*, e68705. **(SCI Q2)**
4. Malche, T., **Tharewal, S.**, & Maheshwary, P. (2024). A TinyML Approach to Real-Time Snoring Detection in Resource-Constrained Wearable Devices. *Engineering Proceedings*, 82(1), 55. **(Scopus Q3)**
5. Malche, T., Maheshwary, P., & **Tharewal, S.** (2024). Instance Segmentation-Based Automated Detection and Thickness Estimation of Oil Spills in Aerial Imagery. *Engineering Proceedings*, 82(1), 6. **(Scopus Q3)**
6. Malche, T., **Tharewal, S.**, & Dhanaraj, R. K. (2023). Automated Damage Detection on Concrete Structures Using Computer Vision and Drone Imagery. *Engineering Proceedings*, 58(1), 60. **(Scopus Q3)**
7. Saju B, Tressa N, Dhanaraj R.K., **Tharewal S**, Mathew J.C., Pelusi D. (2023). Effective multi-class lung disease classification using hybrid feature engineering. *Math Biosci Eng*, 20(11):20245–20273. **(SCI Q2)**
8. Kaur, Veerpal et al., & **Tharewal, S.** (2023). Blockchain + CNN + Hashing for secure 3D biometric storage. *Journal of Discrete Mathematical Sciences and Cryptography*, 26(3), 729–738. **(Scopus Q1)**
9. Singh, Arjun et al., & **Tharewal, S.** (2023). Blockchain-enabled security mechanism for preventing data forgery in IoT-based smart homes. *JDMCS*, 26(5), 1437–1446. **(Scopus Q1)**
10. Navpreet Kaur, Inderdeep Kaur Aulakh, **Sumegh Tharewal**, et al. (2022). Enhanced Route Discovery Mechanism Using Q-Learning. *Scientific Programming*, Article 3187858. **(SCI Q2)**
11. Shobanadevi, A., **Tharewal, S.**, et al. (2022). Novel identity management system using smart blockchain technology. *Int J Syst Assur Eng Manag* 13, 496–505. **(Scopus Q2)**

12. **Sumegh Tharewal**, Mohammed Waseem Ashfaque, and Mohammad Shabaz. Intrusion Detection System for IIoT Using Deep Reinforcement Learning. Wireless Communications and Mobile Computing. **(SCIE Q2)**
13. Timothy Malche, **Sumegh Tharewal**, Pradeep Kumar Tiwari. Intelligent Remote Patient Activity Tracking Using IoT & ML. Journal of Healthcare Engineering. **(SCIE Q2)**
14. **Sumegh Tharewal**, Timothy Malche, Pradeep Kumar Tiwari. Score Level Fusion of 3D Face and 3D Ear for Multimodal Biometrics. Computational Intelligence and Neuroscience. **(SCIE Q1)**
15. Ashish Lahase, Vedpalsing Bhabale, **Sumegh Tharewal**. (2021). A Review on Sentiment Analysis of Hindi and Marathi Scripts. IJARESM, Vol 9, Issue 3.
16. Timothy Malche, Pradeep Kumar Tiwari, **Sumegh Tharewal**, Rajesh Tiwari. (2021). Smart Waste Management System for Smart City using IoT. Turkish J. of Computer & Mathematics Education, Vol.12 No.10. **(Scopus)**
17. Santosh Maher, Shaikh Abdul Hannan, **Sumegh Tharewal**. (2019). HRV-based Heart Disease Prediction using ML. International Journal of Computer Applications, Vol 177, No.27. (UGC care)
18. Santosh K Maher, **Sumegh Tharewal**, Abdul Hannan, Suvarnasing Bhabale, K V Kale. (2018). Review on HRV based Prediction and Detection of Heart Disease. IJCA, Vol 179, No.46. (UGC care)
19. Suvarnasing Bhabale, **Sumegh Tharewal**, Hanumant Gite, Siddharth Dabhade, K.V Kale. (2018). Review on Face, Ear and Signature for Human Identification. IJCA, Vol 180, No.13. (UGC care)
20. Krishna Shinde, **Sumegh Tharewal**. (2017). Development of Face and Signature Fusion Technology. IJERMT, Vol 6, Issue 9. (UGC care)
21. Krishna Shinde, **Sumegh Tharewal**. (2017). Biometrics Technology in Multimode Fusion Data. IOSR-JCE, Vol 19, Issue 5. (UGC care)
22. Kayte Jaypalsing Natthusing, **Sumegh S. Tharewal**, Mohammed Waseem Ashfaque. (2015). Analysis of Cardiac MRI Using Automatic Segmentation Techniques. IOSR-

JVSP, Vol 5, Issue 6. (UGC care)

23. Suvarnsing Bhable, Sangramsing Kayte, **Sumegh Tharewal**, Dr. K.V. Kale. (2015). A Survey on Human Ear Recognition. IJERA, Vol 5, Issue Nov 2015. (UGC care)
24. Kayte Jaypalsing Natthusing, **Sumegh Tharewal**, R. R. Manza. (2014). Analysis of 4D Short Axis Cardiac MRI. Indian Streams Research Journal, Vol 2, Issue 9. (UGC care)
25. Monica Mundada, Sangramsing Kayte, **Sumegh Tharewal**, Bharti Gawali. (2014). Classification of Fluent and Dysfluent Speech Using KNN. IJARCSSE, Vol 4, Issue 9. (UGC care)
26. Jaypalsing Natthusing Kayte, **Sumegh Shrikant Tharewal**, et al. (2014). Left Ventricle Statistical Models for Cardiac MRI. IJCA, Vol 104, No.7. (UGC care)
27. Mohammed Waseem Ashfaq, **Sumegh Tharewal**, et al. (2014). Trends in Education Smart Learning Approach. IJARCSSE, Vol 4, Issue 10. (UGC care)
28. Mohammed Waseem Ashfaq, **Sumegh Tharewal**, Abdul Samad Shaikh, et al. (2014). Text to Voice Conversion for Smart Android Phones. IJEECS, Vol 2, Issue 8. (UGC care)
29. Mohammed Waseem Ashfaq, **Sumegh Tharewal**, Sayyada Sara Banu, Dr. Abdul Hannan. (2014). Implementation & Use of Data Structures in Linux. Vol 4, Issue 1.
30. **Sumegh Shrikant Tharewal**, Mohammed Waseem Ashfaq, Jaypalsing Kayte, et al. (2014). Real-Time BCI System. IJARCSSE, Vol 4, Issue 9. (UGC care)
31. **Sumegh Tharewal**, Sangramsing Kayte, Mohammed W. Ashfaq, Sayyada Sara Banu, Bharti Gawali. (2014). Review of Intelligent Alarm System for Driver Fatigue. IJSER, Vol 5, Issue 9. (UGC care)
32. Nazneen Akhter, Yogesh Rajput, **Sumegh Tharewal**, et al. (2014). Fractals for Complexity Analysis of Diabetic Retinopathy. IJRET, Vol 3, March 2014. (UGC care)
33. Mohammed Waseem Ashfaq, **Sumegh Tharewal**, Abdul Samad Shaikh, et al. (2014). Text Messaging to Voice Message for Smart Android Phone. IJEECS, Vol 2, Issue 9, Nov–Dec 2014. (UGC care)

Books

1. **Dr. Sumegh Tharewal**, Dr. Shantanu Kanade, Dr. Anuradha Kanade, and Dr. Sujeet More, “TERMINOLOGY AND TECHNICAL FOUNDATIONS OF BLOCKCHAINS”, Xoffencer Publication, Published on 1 January 2023,
Amazon Link:- https://www.amazon.in/Terminology-Technical-Foundations-Blockchains-Shrikant/dp/B0C177GKD8/ref=sr_1_72?qid=1682068290&refinements=p_27%3ADr.Anuradha&s=books&sr=1-72
2. Anurag Vijay Agrawal, Mr. P. Prasant, Anurag Prasad, **Dr. Sumegh Shrikant Tharewal**, “FUNDAMENTALS OF BLOCKCHAIN AND CRYPTOCURRENCIES” , Xoffencer Publication, Published on 22 April 2023.
Amazon Link:- https://www.amazon.in/dp/B0C36YTVJJ/ref=sr_1_2?qid=1682069275&refinements=p_27%3ADr.+Sumegh+Shrikant+Tharewal&s=books&sr=1-2
3. Dr. Robin Rohit Vincent, Ms. Vanita Dandhwani , Patrik Viktor , **Dr. Sumegh Shrikant Tharewal**, “ DEVELOPING AND SMART CONTRACTS OF BLOCKCHAIN”
Xoffencer Publication, Published on 1 January 2023.
Amazon Link:- https://www.amazon.in/DEVELOPING-SMART-CONTRACTS-BLOCKCHAIN-Vincent/dp/B0BZ4S9XP7/ref=sr_1_1?qid=1682069275&refinements=p_27%3ADr.+Sumegh+Shrikant+Tharewal&s=books&sr=1-1
4. Dr. Sushil Dohare, Dr. V SelvaKumar, Sachin Raval, **Dr. Sumegh Shrikant Tharewal**, “ INTRODUCTION TO DATA SCIENCE”, Xoffencer Publication, Published on 22 April 2023.
Amazon Link: https://www.amazon.in/INTRODUCTION-DATA-SCIENCE-Sushil-Dohare/dp/B0C1TFFW91/ref=sr_1_3?qid=1682069275&refinements=p_27%3ADr.+Sumegh+Shrikant+Tharewal&s=books&sr=1-3
5. Dr. Araddhana Manisha Arvind Deshmukh, **Dr. Sumegh Shrikant Tharewal**, Dr. Ravinder S Saini, Dr. Bibin K Jose, “ARTIFICIAL INTELLIGENCE APPLICATIONS IN HEALTHCARE”, Xoffencer Publication, Published on 28 April 2023.

Amazon Link: https://www.amazon.in/ARTIFICIAL-INTELLIGENCE-APPLICATIONS-HEALTHCARE-Araddhana/dp/B0C3V4C9PD/ref=sr_1_2?qid=1683013404&refinements=p_27%3ADr.+Sumeegh+Shrikant+Tharewal&s=books&sr=1-2

6. Dwivedi, U., Shukla, S. K., & **Tharewal, S.** (2025, June 16). Data Decisions Unlocked: Redefining Intelligence with Next-Gen Big Data Thinking. Xilvora Ink Pvt. Ltd. Retrieved from <https://xilvoraink.com/data-decisions-unlocked/>

Book Chapters

1. Awasthi, Y., Singh, S., Rai, S., Maheshwari, R., Upreti, K., & **Tharewal, S.** (2025). Maximizing investment potential: A comparative study of Nifty 50 and Nifty Midcap 100 using advanced machine learning-based financial models. In BookSmart Cyber-Physical Systems (Chap. 9). CRC Press. <https://doi.org/10.1201/9781003542513-9> (**Scopus**)
2. Balan, A., Tomar, R., Ekature, S. R., Upreti, S., **Tharewal, S.**, & Poonia, R. C. (2025). Transforming autonomous vehicle technology: AI and machine learning applications in cyber-physical systems. In Smart Cyber-Physical Systems: Innovations and Real-World Implications (Chap. 11). CRC Press. <https://doi.org/10.1201/9781003542513-11>(**Scopus**)
3. Singh, N., Upreti, S., Upreti, K., **Tharewal, S.**, & Kumar, N. (2025). Exploring the impact of smart watches on health management for senior citizens: A qualitative study. In Big Data Analytics and Intelligent Applications for Smart and Secure Healthcare Services (Chap. 10). CRC Press. <https://doi.org/10.1201/9781003508618-10>(**Scopus**)
4. Kaur, V., Bhatt, D. P., **Tharewal, S.**, & Tiwari, P. K. (2024). Blockchain-Based Cyber-Physical System: Opportunities and Challenges. In Intelligent Security Solutions for Cyber-Physical Systems (pp. 1–12). Chapman & Hall/CRC. <https://doi.org/10.1201/9781003406105-20>(**Scopus**)
5. Malik, K., Singh, O., & **Tharewal, S.** (2024). Integrating telemedicine into population health management for pharmacogenomics and patient-centered care in rural India. In Convergence of Population Health Management, Pharmacogenomics, and Patient-Centered Care (Chap. 8). IGI Global. <https://doi.org/10.4018/978-8-3693-3260-3.ch008>(**Scopus**)

6. Nagsen Bansod, Siddharth Dabhade, Santosh Maher, **Sumegh Tharewal**, K. V. Kale. (2019). Human Electroencephalographic Biometric Person Recognition System. Lecture Notes in Networks and Systems, Springer. (**Scopus**)
7. N. Akhter, **S. Tharewal**, V. Kale, A. Bhalerao, & K. V. Kale. (2016). Heart-Based Biometrics and Possible Use of Heart Rate Variability in Biometric Recognition Systems. In Advanced Computing and Systems for Security, Springer India, pp. 15–29. DOI: 10.1007/978-81-322-2650-5_2 (**Scopus**)

Research Paper Accepted

1. Veerpal Kaur, Devershi Pallavi Bhatt, **Sumegh Tharewal**, Pradeep Kumar Tiwari, “ Intelligent Multimodal 3D Biometric Recognition using PointNet++ for Robust Face–Ear Authentication”, Scientific Reports by Nature(**SCI Q1, Impact Factor 4.3**)
2. Timothy Malche, **Sumegh Tharewal**, Rajesh Kumar Dhanraj, Priti Maheshwary , Adya Tiwari , Dragan Pamucar, “TinyML-Enabled Smart Jacket for Fall Detection and Human Activity Recognition using Internet of Things (IoT) ”, Internet of Things; Engineering Cyber Physical Human Systems. (**SCI Q1, impact factor 5.9**)
3. Binju Saju, **Sumegh Tharewal**, Kamal Upreti, Rituraj Jain, Uma Shankar, Arpana Prasad, Neethu Tressa, “HNRET- A Novel Hybrid Deep Learning Based Model for Noise Reduction and Enhancement for CT Scan Kidney Images”, Menoufia Journal of Electronic Engineering Research (**Scopus Q3**)
4. Rathore, Yogesh Kumar; **Tharewal, Sumegh**; Upreti, Kamal; Sivaneasan, B.; Kshirsagar, Pravin; Tran, Tien Anh, “Intelligent Modeling of Charging Behavior for Electric Vehicle Batteries under Varying Thermal Conditions”, Engineering Reports by wiley (**Scopus Q2, impact factor 2**)

Research Paper Communicated

1. S. Gomathi, **Sumegh Tharewal**, Sumit Kumar, Shaikh Abdul Hannan, Shakir Khan, Melkamu Tsegaye, “Lightweight Explainable machine learning and orange data mining tool for Covid-19 prediction”, Computational Intelligence and Neuroscience by Hindawi Publishers. (**SCIE, Q1, Impact factor 3.633**)
2. Krishna Shinde, **Sumegh Tharewal**, Charansing Kayte, “Face Recognition for Person Identification based on Deep Learning Techniques”, Computational Intelligence and

Neuroscience by Hindawi Publishers. (**SCIE, Q1, Impact factor 3.633**)

3. Bharat Naiknaware, **Sumegh Tharewal**, Ashish Bhalerao, “User’s opinion on Covid-19 Vaccination using Machine Learning and Lexicon Based Approach”, Computational and Mathematical Methods in Medicine. (**SCIE, Q2, Impact factor 2.238**)
4. **Sumegh Tharewal**, Timothy Malche, Emanuele Caruso, Oswald Lanz, “Real-Time 3D Human Pose Estimation and Action Tracking for Enhanced Human–Robot Collaboration Using MediaPipe and MMPose”, Machine Vision and Applications (**Scopus Q2**)

Skills

✚ Machine Learning, Artificial Intelligence, Deep Learning, Digital Image Processing, Biometrics, Pattern Recognition, Computer Vision, Open CV, Internet of Things

Programming Languages

✚ C, C++, Python, MATLAB, HTML, JAVA, PHP, visual basic, .NET, PL/SQL, Oracle

Certificate and Badge

- 1] Python for Data Science by **IBM**.
- 2] Digital Skills: Artificial Intelligence by **Accenture**.



- 3] Digital Skills: Digital Marketing by **Accenture**.
- 4] Machine Learning with Python by **IBM**.



- 5] Machine Learning onramp by Mathworks Training Services.
- 6] Deep learning onramp by Mathworks Training Services.
- 7] Artificial Intelligence Concepts by IBM Training.
- 8] Introduction to Artificial Intelligence by IBM and Coursera.

Startup Certification

- ✚ **Startup India Learning Program**, by Startup India and upGrad under the Invest India National Investment Promotion Agency, July 10, 2020. Certificate no: - 20473962

Student Club formation

- ✚ Club Name: - **Google Developer Club**

Taken Initiative and established at MIT WPU

Role and Responsibility: - Working as a faculty advisor and mentor

Budget Approved: - **4 lakh per annum.**

- ✚ Club Name: - **Chanakya Coding Club**

Taken Initiative and established at MIT WPU

Role and Responsibility: - Working as a mentor.

- ✚ Club Name: - **Innoventures Entrepreneurship Club**

Taken Initiative and established at DBS Global University, Dehradun.

Role and Responsibility: - Working as a Patrons

Leadership Program

- ✚ Participated and completed Five Days Residential Programme on Nurturing Future Leadership Programme (NFLP) under the Malaviya Mission Teacher Training Programme (MMTTP) scheduled to be held from 28th June to 2nd July 2025 at IIT Roorkee.

Faculty Development Program

1. Participated and completed Five Days Faculty Development Program on Blockchain Technology, organized by Dr. Vishwanath Karad MIT World Peace University, Pune, in association with MIT-WPU CSI Chapter, from 18–23 April 2022.

2. Participated and completed Five Days National Level E-FDP cum Workshop on Emerging Computational Technologies, organized by the Department of Computer Applications, Manipal University Jaipur, from 10–14 October 2022.
3. Participated in the Five-Day Faculty Development Workshop on Redefining Innovative Teaching–Learning for Academic Improvisation, organized by MUJ-TEC, Manipal University Jaipur, from 15–19 February 2021.
4. Participated in the Five-Day Faculty Development Program on Technology Enabled Teaching and Learning, organized by Manipal University Jaipur, from 2–6 August 2021.
5. Participated and completed AICTE Training and Learning (ATAL) Academy Online Elementary Five-Day FDP on Artificial Intelligence, from 20–24 September 2021, organized by Dr. Babasaheb Ambedkar Marathwada University, Aurangabad.
6. Participated in the Seven-Day International FDP on “Innovation in Information Technology Post COVID-19”, organized by the UG Department of Computer Applications (BCA) in association with the Computer Society of India (CSI – NGM Student Branch), from 21–27 May 2020.

Workshop/Seminar Attended

1. Participated in Expert Lecture Series on Intellectual Property Right–2021 titled “How to commercialize your IPR” on 10 March 2021, organized by Manipal University Jaipur.
2. Participated in Expert Lecture Series on Intellectual Property Right–2021 titled “Creativity in fashion and art world and its legal protection” on 9 March 2021, organized by Directorate of Research, Manipal University Jaipur.
3. Participated in Expert Lecture Series on Intellectual Property Right–2021 titled “IPR for Universities and Technical Institutes” on 8 March 2021, organized by Directorate of Research, Manipal University Jaipur.
4. Participated in International Conference on Applied Scientific Computation Intelligence (ASCI-2020) on 22–23 December 2020, organized by Department of Computer Application, Manipal University Jaipur.
5. Participated in International Symposium: Recent Advancement in ICT on 29 December 2020, organized by CSI Aurangabad Chapter.

6. Participated in Machine Learning for Remote Sensing Data Classification Workshop on 1 June 2020, organized by IIRS Outreach, ISRO Dehradun.
7. Participated in the IEEE Bangalore Section Researchers Forum Online Workshop on “IEEE Reference Management Tool for Research Articles and Thesis Writing” on 20 May 2020.
8. Participated in International Webinar on Writing Articles in Impacted Journals on 19 May 2020, organized by MGM University, Dr. G. Y. Pathrikar College of CS & IT, Aurangabad.
9. Participated in Virtual National Symposium on Soft Biometrics on 18 May 2020, organized by MGM University, Dr. G. Y. Pathrikar College of CS & IT, Aurangabad.
10. Attended One-Day Online Workshop on Crowd Sourcing and Participatory-GIS, organized by IIRS Outreach, ISRO Dehradun (2020).
11. Participated in One-Day Online Workshop on Satellite Data Access from Online Repositories and ISRO Bhuvan Geoportal on 25 June 2019, organized by IIRS Outreach, ISRO Dehradun.
12. Attended One-Day Scilab Workshop on 4 May 2019, organized by IIT Bombay TLC (ICT) and MGM’s Dr. G. Y. Pathrikar College of CS & IT, Aurangabad.
13. Participated in One-Day Seminar on Intellectual Property Rights on 6 February 2019, organized by European IPR Helpdesk and MGM’s Dr. G. Y. Pathrikar College of CS & IT, Aurangabad.
14. Participated Two Days Zonal Level Competition ANVESHAN at Ganpat University, Mehsana, Gujarat on 14–15 February 2018.
15. Participated in One-Week GIAN Course on “Perception and Molding of Three-Dimensional Scenes” under MHRD (GIAN Phase II), from 4–10 December 2017, organized by Dept. of CS & IT, Dr. B.A.M. University Aurangabad.
16. Participated in Author Workshop on Scientific Writing & Publishing Scholarly Articles on 7 November 2017, at Knowledge Resource Center, Dr. B.A.M. University Aurangabad.
17. Participated in the 104th Indian Science Congress at Sri Venkateshwara University, Tirupati from 3–7 January 2017.

18. Participated in One-Day Seminar on Cyber Law and Cyber Security on 18 January 2017, organized by Dr. G. Y. Pathrikar College of CS & IT, Aurangabad.
19. Participated in Two-Day National Workshop on Multicore Technology from 17–18 February 2016, organized by IETE Aurangabad and MGM's Dr. G. Y. Pathrikar College of CS & IT.
20. Participated in One-Day Skill & Knowledge Enhancement Program (SKEP) by IEEE Bombay Section and MGM, Dr. G. Y. Pathrikar College of CS & IT, on 5–6 August 2016.
21. Participated in the 103rd Indian Science Congress at University of Mysore from 3–7 January 2016.
22. Participated in University–Industry Interaction Summit 2015 at Dr. B.A.M. University, Aurangabad from 29–30 October 2015.
23. Participated in the 102nd Indian Science Congress at University of Mumbai from 3–7 January 2015.
24. Participated in Science Day Speech Competition on 28 February 2014, Department of CS & IT, on National Science Day.
25. Participated in Two-Day National Workshop on Microwave Remote Sensing from 22–24 January 2014, organized by Srinivasa Ramanujan Geospatial Chair, Dr. B.A.M. University.
26. Participated in One-Day Seminar on MATLAB & Simulink for Engineering Education on 7 November 2014, organized by MathWorks India at Hotel Lemon Tree, Aurangabad.
27. Participated in One-Day State Level Seminar on Digital Signal and Image Processing (DSIP-2013) on 15 October 2013, organized by SRTM University Sub-Center Latur.
28. Participated in One-Day Workshop on Image Processing Techniques using MATLAB 2012 on 24 February 2013, organized by Dept. of Computer Science, Maulana Azad College Aurangabad and Ajanta Prakashan.

Webinar Attended

1. Webinar attended on “Vision: Marathwada 2025: Towards Atmanirbhar Bharat”, organized by AIC-BAMU Foundation and Rubiscape Pvt. Ltd. on 29 August 2020.
2. Webinar attended on Artificial Intelligence with Er. Karan Arora (Founder & CEO, Itronix Solutions) on 11 July 2020, organized by Itronix Solutions.

3. Participated and successfully completed a webinar on “Cyber Security” on 9 July 2020, organized by Department of Computer Science, Shri Shivaji Science & Arts College, Chikhli.
4. Participated and successfully attended Leadership Talk with Mr. Shridhar Vekant (CEO, Akshay Patra Foundation) on 20 June 2020, organized by MHRD’s Innovation Cell, Government of India.
5. Participated and successfully attended Leadership Talk with Ms. Ashwini Deshpande (Co-Founder & Director, Elephant Design) on 6 June 2020, organized by MHRD’s Innovation Cell, Government of India.
6. Participated in International Webinar Series on “Forensic Science for Society” from 25 May – 02 June 2020, organized by G.T.N. Arts College (Autonomous), Tamil Nadu.
7. Participated in Webinar on “AI and Data Science” on 30 May 2020, organized by IEEE Circuits and Systems (CAS) Society Bangalore Chapter with IEEE Bangalore Section.
8. Participated in International Web Conference on “Impact of COVID-19 on Education System” on 30 May 2020, organized by Dr. Babasaheb Ambedkar Marathwada University Aurangabad in association with CSI and IETE.
9. Participated in Webinar on “Impact of Blockchain” on 29 May 2020, organized by MGM University, Dr. G.Y. Pathrikar College of CS & IT, Aurangabad.
10. Participated in Webinar on “The Journey of Data Science” on 26 May 2020, organized by MGM University, Dr. G.Y. Pathrikar College of CS & IT, Aurangabad.
11. Participated in online Webinar on “Googling Google” on 26 May 2020, organized by Department of CSE, S.B.M. Polytechnic, Mumbai.
12. Participated and successfully completed Webinar Series on “Machine Learning and Artificial Intelligence” from 21–27 May 2020, organized by IEEE Bombay Section and SPIT Mumbai.
13. Participated in State Level Webinar on “Blockchain Technology” on 20 May 2020, organized by Chalisgaon Education Society’s B.P. Arts, S.M.A. Science & K.K.C. Commerce College, Chalisgaon.
14. Participated in webinar on “Mental Health Effects of the COVID-19 Pandemic” in May 2020, organized by Harvard Medical School, Boston, USA.

Invited as a Resource Person

- ✚ Invited as a **Resource Person** for the upcoming **AICTE ATAL Faculty Development Programme (FDP) “Next-Gen Computing for Sustainability: Intelligent Technologies and Data-Driven Solutions”**, organized by CHRIST (Deemed to be University), Delhi NCR, Ghaziabad, India, from 17th November 2025 to 22th November 2025.
- ✚ Invited as Session Expert for the International Short-Term Training Program (STTP) on **“AI-Powered Cyber Threats and Defenses”**, held from 16–21 June 2025 at Symbiosis Skills and Professional University, Pune (online mode).
- ✚ Invited as Resource Person for the **Ignite and Inspire Program (Career Guidance and Development)** at Deogiri College, Chhatrapati Sambhajinagar on 21 March 2025.
- ✚ Invited as Resource Person for a **One-Day Workshop on Emerging Trends in Machine Learning** at Yeshwantrao Chavan College of Arts, Commerce & Science, Sillod, on 20 March 2025.
- ✚ Invited as Resource Person for a research talk on **“Emerging Trends in Artificial Intelligence”** at Dr. G. Y. Pathrikar College of CS & IT, MGM University, Chhatrapati Sambhajinagar, on 19 March 2025.
- ✚ Invited as Resource Person for a **Two-Day Workshop on Startup and Entrepreneurship** at Shankarlal Khandelwal College, Akola, under the PM-Usha Central Government Scheme, on 17–18 March 2025.
- ✚ Invited as Resource Person for **SWARM Presents: Computer Vision in AI, AI Forum USA** (online mode), on 3 October 2024.
- ✚ Invited as Resource Person for a webinar on National Innovation Day on the topic **“Creativity, Innovation and Entrepreneurship”**, organized by the Department of Forensic Science and Institution’s Innovation Council, G.T.N. Arts College (Autonomous), Tamil Nadu.
- ✚ Invited as Resource Person for the IETE Aurangabad Centre – Lecture Series on **“Current Trends in Multimodal Biometrics”**, organized by IETE Aurangabad Centre on 28 August 2021.
- ✚ Invited as Resource Person for a 6-Day Online FDP on **“Research Opportunities and Challenges in Multimodal Biometrics”**, organized by the Department of CS & IT and the Department of Animation, Deogiri College Aurangabad, on 11 August 2021.

- ✚ Invited as Resource Person for a 5-Day Online FDP on “**Recent Advances in Multimodal Biometrics**”, organized by the Department of ECE in association with IIC, IETE, and IEDC Cells, IES College of Engineering, Thrissur, on 21 June 2020.
- ✚ Delivered a session on “**Recent Advancement in Biometrics**” at Deogiri Institute of Engineering and Management Studies, Aurangabad, on 20 February 2019.
- ✚ Invited as Speaker for a **National Level Seminar on Artificial Intelligence** at S.D. Deshmukh College, Bhokardan, District Jalna, on 22 February 2019.
- ✚ Invited as Speaker for a **National Level Workshop on Machine Learning** at S.D. Deshmukh College, Bhokardan, District Jalna, on 24 February 2019.
- ✚ Invited as Speaker for a **National Level Workshop on Multimodal Biometrics** at S.D. Deshmukh College, Bhokardan, District Jalna, on 24 January 2019.
- ✚ Invited as Speaker for a session on “**Multimodal Biometric System**” at Deogiri Institute of Engineering and Management Studies, Aurangabad, on 30 January 2018.
- ✚ Delivered a talk on “**Multimodal Biometrics and Its Applications**” at the Government Institute of Forensic Science, Aurangabad, on 20 March 2018.

Invited as a Keynote Speaker

- ✚ Invited as **Keynote Speaker** in International Conference (International Conference on Smart Cyber Physical Systems (ICSCPS-2024) at Christ University Delhi NCR (<https://icscps.christuniversity.in/icscps-24/?page=speakers>) September 13 - 14, 2024.
- ✚ Invited as **Keynote Speaker** in International Conference at Samarkand University, Uzbekistan (The 5th International Conference on Trends in Electronics and Health Informatics (TEHI 2025) 7–8 November 2025 (<https://www.tehi2025.uz/session>)
- ✚ Invited as a **Keynote Speaker** in The Global AI Summit 2025, with the theme "International Conference on Artificial Intelligence and Emerging Technology 2.0 (AISUMMIT-2025)," from November 19th to 21th, 2025, Technically sponsored by IEEE UP section hosted by Bennett University, Greater Noida, Uttar Pradesh, India
- ✚ Invited as Keynote **Speaker** in NICEDT-2026: NIELIT's International Conference on Communication, Electronics and Digital Technologies, 29-30 January 2026, Gorakhpur, India

Reviewer of Journal

- ✚ Reviewer of IEEE Transactions on Consumer Electronics
- ✚ Reviewer of Artificial Intelligence by MDPI
- ✚ Reviewer of World Journal of Engineering by Emerald Publishing. Scopus Indexed Journal ISSN: 1708-5284.
- ✚ Reviewer of International Journal of Systems Assurance Engineering and Management (IJSA) by Springer Nature.
- ✚ Reviewer of SN Computer Science by Springer Nature.
- ✚ Reviewer of Pattern Recognition Letters by Elsevier
- ✚ Reviewer of Artificial Intelligence by Elsevier

Online Examination Experience

1. Worked as **Dy Center Coordinator in DRDO CEPTAM** (Centre for Personnel Talent Management Exam), in November 2019.
2. Worked as **Center Coordinator in DRDO CEPTAM** (Centre for Personnel Talent Management Exam), December 2020.
3. Worked as **Observer in National Testing Agency NTA** (An autonomous organization under the Department of Higher Education, Ministry of Education, and Government of India) NEET (UG) - 2021 to be held on 12 September 2021.

Proposal Sanctioned for Start-up

- ✚ Selected proposal for start-up entitled “Low-Cost 3D Scanner” at **Bajaj Incubation Center** Dr. Babasaheb Ambedkar Marathwada University Aurangabad (MH) India.

Experience as a Session Chair

- ✚ Chaired the Session at National Conference on Emerging Trends in Engineering (NCETE- 2020) on 14 June 2020 Organized by Geetanjali Institute of Science & Technology Gangavaram(V), Kovur(M), Nellore(D), Andhra Pradesh, India in association with InSc.
- ✚ Chaired the Session in 3rd National Students' Research Conference to be held on 28th Feb. 2022, organized by School of Computer Science Dr. Vishwanath Karad MIT-World Peace University, Pune.

- ✚ Chaired the session in the 2nd International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIHI-2025) will be held on December 4–5, 2025, at Shri Shankaracharya Institute of Professional Management and Technology (SSIPMT), Raipur, Chhattisgarh, India.
- ✚ Chaired the session in the 5th International Conference on SMART SYSTEMS: INNOVATIONS IN COMPUTING (SSIC-2025), organized by the Department of Information Technology, Manipal University Jaipur December 19-20, 2025.

Experience as a Session Coordinator

1. Worked as a session Coordinator in International Conference on Applied Scientific Computational Intelligence using Data Science (ASCI-2020), December 22-23, 2020, Organized by Manipal University Jaipur.
2. Worked as a session Coordinator at 3rd International Conference on Smart Systems: Innovations in Computing 22- 23 Jan 2021, Organized by Manipal University Jaipur.
3. Worked as a session coordinator in International Conference on Digital Resilience for Ecological Advancement, Management and Sustainability (DREAMS) 22-23 August 2025.

Experience as a Technical Program Committee (TPC) Member

1. Worked as Technical Program Committee (TPC) member in the 3rd International Conference on Soft Computing and its Engineering Applications (icSoftComp2021) organized by Smt. Chandaben Mohanbhai Patel Institute of Computer Applications, Charotar University of Science & Technology (CHARUSAT), Changa, India during December 10-11, 2021.
2. Technical Program Committee (TPC) member of 4th International Conference on Soft Computing and its Engineering Applications (icSoftComp2022) during December 9-10, 2022.
3. Technical Program Committee Member of First International Conference on Advances in Computer Vision and Information Technology (ACVAIT 2022) organized by Department of Computer Science and Information Technology, Dr. Babasaheb Ambedkar Marathwada University, Aurangabad in Collaboration with Curtin University Malaysia, Rani Channama

University Belgavi, SRTM University Nanded, Dr. Babasaheb Ambedkar Technological University (BATU) Lonere, SNDT Womans University Mumbai, MGM University Aurangabad and Vivekanand College Aurangabad, 1-2 August 2022.

4. Technical Program Committee Member of 3rd International Conference on Computing, Analytics and Networks (ICAN2022), organized by Chitkara University, Rajpura, Punjab, India, 18-19, November 2022.
5. Technical Program Committee Member of the 5th International Conference on Recent Trends in Image Processing & Pattern Recognition (RTIP2R) December 1-2, 2022, Texas A&M University - Kingsville, Texas (USA). in collaboration with 2AI: Applied AI Research Lab, Computer Science, USD (USA)
6. Technical Programme Chair at the 2nd International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI-2025), held on 4th–5th December 2025.

Experience as a Coordinator

- ✚ Innovation Conclave of MIT World Peace University, Research Innovation Design Entrepreneurship (RIDE22), 20- 25 September 2022.

Experience as a Convener

- ✚ Convener for the International Conference on Digital Resilience for Ecological Advancement, Management and Sustainability (DREAMS 2025), hosted by DBS Global University, Dehradun, held on 22-23 August 2025.
- ✚ Convener of 6-Days International e-FDP On "Applications of Machine Learning in Internet of Things (IoT) and Data Analytics" from 19-June-2023 to 26-June-2023, symbiosis Institute of Computer Studies and Research (SICSR), Symbiosis International (Deemed) University (SIU), Pune
- ✚ Convener of 3rd National Students' Research Conference, 28 Feb 2022, MIT World Peace University, Pune.
- ✚ Convener of Expert Talk on “How to Create a Successful Startup”, organized by Department of Computer Applications in association with Directorate of E-cell and Atal Incubation Centre Manipal University Jaipur, September 13, 2021.

- ✚ Convener of Research Expert Talk on “Effective Research Paper and Publication organized by Department of Computer Applications Manipal University Jaipur, December 1, 2021.
- ✚ Convener of Industry Expert Talk on “Kickstart Web Development using ReactJS” organized by Department of Computer Applications Manipal University Jaipur, December 8, 2021.
- ✚ Convener of Industry Expert Talk on “Role and Demand of DBA in IT Industry”, organized by the Department of Computer Applications Manipal University Jaipur, December 16, 2021

Active Professional Membership

- ✚ IEEE membership number 93189911
- ✚ Life member of Indian Science Congress membership number L26426
- ✚ IAENG (International Association of Engineers)
- ✚ IACSIT (International Association of Computer science & Information Technology)
- ✚ EURASIP (European Association of Signal Processing)
- ✚ IEEE Biometric Council
- ✚ Face Recognition Research Community
- ✚ Institute of Scholars Bangalore (Insc2019761)
- ✚ iBridge Internal Innovation Management at Organizations.
- ✚ International Multidisciplinary Research Foundation (IMRF) Registered with Government of India NITI Aayog NGO Darpan, RSquareL

Languages

- ✚ **English:** - Professional Speaker
- ✚ **German:** - A1.1
- ✚ **Hindi:** - Native Speaker
- ✚ **Marathi:** - Native Speaker

Personal Information

- ✚ **LinkedIn:** - <https://www.linkedin.com/in/sumegh-tharewal-30bb4767/>
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Name and addresses of three References

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Declaration

The above information is correct to the best of my knowledge.

Dr. Sumegh S Tharewal